Jonathan Yu

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Education

University of California, Berkeley **Expected May 2028**

B.S. Chemistry, B.A. Computer Science; GPA: 3.92; SAT: 1580

Relevant Courses: Deep Learning, Theoretical Statistics, Optimization, Probability, Algorithms, Data Science, Organic Chemistry, Physical Chemistry

Research Experience

Incoming Venture Development Analyst, Confidential Life Sciences Venture *New York, NY* **June 2027**

Incoming Discovery Fellow for Agentic Drug Optimization, Merck *UC Berkeley and Merck–Berkeley, CA* **September 2026**

Researcher, Toste Group *UC Berkeley and Center for Computer Assisted Synthesis–Berkeley, CA* **July 2025 – Present**

- Build multimodal data ingestion/annotation pipelines for vision-language models to infer reactions from primary literature
- Curate large-scale, chemically diverse reaction datasets emphasizing wet-lab validation and underrepresented transformations
- Identify coverage gaps in existing template- and agent-based approaches to downstream reaction prediction/drug optimization

SURF Fellow, Zhang Lab *NSF Center for Computer Assisted Synthesis–Notre Dame, IN* **June 2025 – Present**

- Built agentic chemical reasoning harnesses grounding hypothesis generation with literature-derived graph as process verifier
- Developed ReAct and Tree-of-Thought prompting workflows to evaluate long-horizon chemical reasoning in large language models
- Analyzed failure modes in multi-step synthesis planning between structured search/reasoning strategies and 0-shot baselines
- Presented findings to C-CAS, identifying verifier-interface failures under exposed reasoning traces for chemistry benchmarks

Researcher, Gong Lab *UC Berkeley, School of Optometry–Berkeley, CA* **January 2025 – June 2026**

- Co-lead in silico analysis of cataract-associated Cx50-S50P gap junctions to characterize disease-associated functional disruption
- Construct membrane-embedded connexin assemblies and simulate mutation-induced pore constriction dynamics in GROMACS
- Predict intrinsically disordered Periaxin variant structures *de novo* using AlphaFold3 and Rosetta3-based fragment construction
- Analyze transcriptional dynamics in lens development in 1,000+ genes/15+ transcription factors; identified Hes1 as key regulator
- Maintain 500+ mice with critical genotype modifications; extracted lens tissue for proteomic and transcriptomic profiling

Rockefeller SSRP Intern, Fuchs Lab *The Rockefeller University – New York, NY* **June 2024 – August 2024**

- Designed and performed cell stressor/cytoprotector perturbation protocol for keratinocytes and squamous cell carcinomas
- Achieved 20% decrease in cell proliferation with psoriasis-associated interleukin perturbations in presence of retinoic acid
- Presented at Rockefeller Summer Poster Session; highlighted regulatory mechanisms in skin barrier maintenance

UCSC SIP Intern, MacMillan Lab *UCSC, Department of Chemistry and Biochemistry – Santa Cruz, CA* **June 2023 – August 2023**

- Cultured and extracted novel non-enzymatic natural products from *Erythrobacter* primed with fluorinated anthranilic acid scaffold
- Interpreted LCMS fragmentation patterns to infer reactivity and structural motifs; validated structure via COSY/HSQC/HMBC NMR
- Presented at UCSC SIP Research Session, showcasing reactivity-based natural product synthesis with one final lead molecule

SSP Participant at Indiana University *The Summer Science Program – Bloomington, IN* **June 2022 – July 2022**

- Identified PdCdc14 as a key target in preventing citrus crop rot by inhibiting its mitotic phosphatase function in *P. Digitatum*
- Computationally generated homologous PdCdc14 scaffolds; verified ortholog structure conservation using sequence alignment
- Scored docking affinities of 12 FDA-approved small-molecule scaffolds and 40+ respective derivative candidates in CCG MOE
- Submitted findings on catalytic kinetics, inhibitor candidates, and active site sterics to *Purdue University Research Repository*

Leadership Experience

Internal Vice President, Pride Business Association *UC Berkeley–Berkeley, CA* **Fall 2025–Present**

- Lead Berkeley Haas's only queer undergrad business organization, building a 25+ member community for queer students

Internal Vice President, Out in STEM *UC Berkeley–Berkeley, CA* **Spring 2026 – Present**

- Managed $4,000+ in corporate sponsorships, coordinating external partners and securing funding for 25+ student community

Catalysis Team Lead, Biofuels Technology Club *UC Berkeley–Berkeley, CA* **Fall 2024 – Present**

- Lead team of 3 students to increase biofuel yields by 20% while decreasing solvent use by 50% at a laboratory scale

Finance and Peptide Design Project Lead, iGEM at Berkeley *UC Berkeley–Berkeley, CA* **January 2025 – June 2026**

- Secured $2,000+ in funding and organized 400+ attendee biotech symposium across 10+ industry and academic partners

Skills & Interests

Skills: Python, SQL, PyTorch, XGBoost, Linux/HPC, pandas, numpy, R, git

Modeling & Tools: Rosetta Suite/RFDiffusion, GROMACS, CHARMM, AlphaFold, PyMOL, CCG MOE, MNova, Xcalibur, ImageJ

Interests: ChemML, traveling, aviation, skiing, fragrance, floriculture, zinemaking, food science, mouseless Excel, *Gossip Girl*